

CS 515: Advanced Computer Science Practicum

Course Code	CS 515
Course Name	Advanced Computer Science Practicum
L-T-P-C	2-0-2-3
Prerequisites	UG level courses on Computer Networks, Operating Systems, Databases, Computer Organization and Compilers
Intended for	MTech (CSE), MS, PhD
Distribution	Discipline Core for MTech CSE, Elective for others
Approval	recommended by the 40th BoA (May 2021)

Course Contents

- **Databases:** Assignment on implementing an end-to-end system involving database backend, front-end web-based visualization and using database queries, joins, procedures, triggers, views, cursors, and transactions. Possible problems include: [8 Lectures]
 - Create a backend database application with a front-end web-based visualization that plots dependent variables of interest.
 - Create a front-end website with the ability to register and authenticate users with respect to usernames and passwords; write a trigger on a table in a database that allows us to back-up the deleted entries in the table; write other complex queries involving joins on multiple tables.
- **Operating Systems:** Assignments on understanding practical aspects of synchronization and implementing kernel modules in Linux. Modules allow extending the function of the kernel without rebooting the system. This assignment will introduce the students to implementing simple modules, their compilation and also tools for debugging kernels. [8 Lectures]
- **Computer Networks:** Assignments that involve socket programming, and protocol analysis by using network simulators. [8 Lectures]
- **Compilers and Architecture:** Assignments that involve understanding the mapping of a high-level program down to a machine. The idea would be to familiarize students with the nuances of the connection between translating a high-level program and the machine on which it is executed. The assignments could be a subset of: [16 Lectures]
 - Writing a translator for converting programs written in a small programming language to an intermediate representation.
 - Writing an interpreter for evaluating expressions/statements in a (small) programming language.
 - Writing a register allocator for a given reference architecture.
 - Writing an assembler that translates programs in an intermediate representation to the assembly for a given architecture.
- Mini-project on designing scalable systems for different applications. This project will follow software engineering principles of project management, will emphasize scalable design principles, and involve performance analysis. [16 hours]

References

1. Jonathan Corbet, Alessandro Rubini, and Greg Kroah-Hartman, **Linux Device Drivers**, 3rd Edition, O'Reilly Media, 2005.
2. Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, **Operating Systems: Three Easy Pieces** by Arpaci-Dusseau Books, 2018.
3. Kurose and Ross, **Computer Networking: A Top-Down Approach**, 6th Edition, Pearson 2013.
4. Andrew W. Appel and Jens Palsberg, **Modern Compiler Implementation in Java**, 2nd Edition, Cambridge University Press, 2002.
5. Elmasri, R., & Navathe, S. B., **Fundamentals of Database Systems**, 7th Edition, Pearson Education, UK, 2015.

This course is proposed as a core course for MTech (CSE) and BTech students will not be allowed to credit the same. The overlapping contents are essentially with various BTech core courses, and thus the intended audience is completely different.